

Fine-tuning Assignment Report

AI Browser Extension for Code Generation

Home Assignment:

Start the fine-tuning process step by step by posting each stage in the group. After receiving approval, proceed with starting the fine-tuning job. Once the fine-tuned model is successfully created, integrate it into the AI extension codebase inside the **popup.js** file and test the code generation flow.

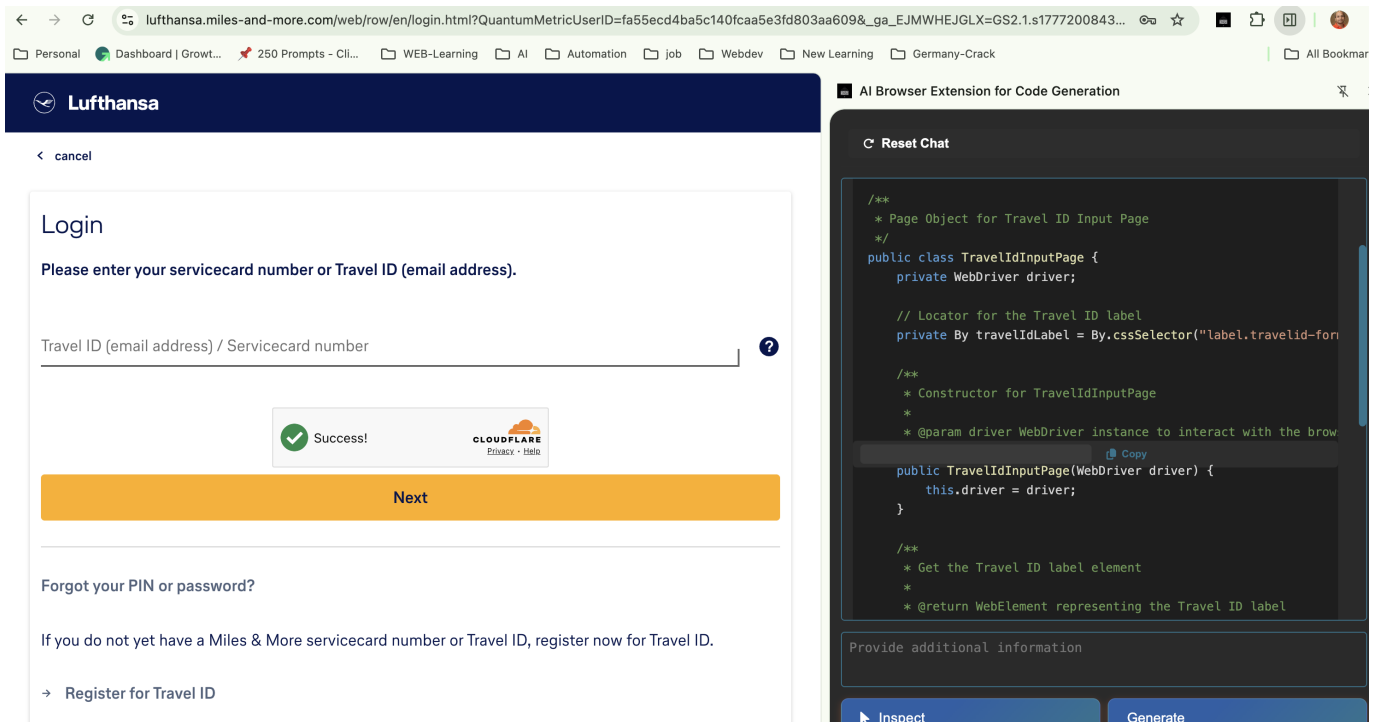
Item	Details
Assignment Goal	Fine-tune a model and integrate it into the browser extension code-generation workflow.
Base Model	gpt-4o-mini-2024-07-18
Output Model	ft:gpt-4o-mini-2024-07-18:testleaf-2::DkrC1OVG
Training Method	Supervised fine-tuning
Training Status	Succeeded
Created At	29 May 2026 18:27:42
Integration File	src/scripts/popup.js
Test Evidence	Model used through the extension to generate Selenium Java Page Object code.

Completed Work

- 1. Prepared training data** - Converted existing page-object examples into JSONL chat-format training records using the training-data converter.
- 2. Started supervised fine-tuning** - Submitted the training file and ran the fine-tuning job using the selected OpenAI base model.
- 3. Verified fine-tuning completion** - Confirmed that the fine-tuning job succeeded and captured the output model details.
- 4. Integrated output model** - Added the fine-tuned model option into popup.js under the Testleaf provider list.
- 5. Tested code generation flow** - Opened a live web page, inspected the UI, selected the trained model, and generated Selenium Java Page Object code from the extension.

Evidence Screenshots

Evidence 1



Testing the fine-tuned model in the AI Browser Extension on the Lufthansa login page. The extension generated a Selenium Java Page Object from the inspected web page.

Evidence 2

Fine-tuning

Train and manage custom models

1

TOTAL

All

Succeeded

FINE-TUNE ID

ft:gpt-4o-mini-

ft:gpt-4o-mini-2024-07-18:testleaf-2::DkrC1OVG

×

✓ Status

Succeeded

🔍 Job ID

ftjob-autBm2QZSn3XHgjEwmkSXvkF

📄

🔍 Training Method

supervised

📦 Base Model

gpt-4o-mini-2024-07-18

📦 Output Model

ft:gpt-4o-mini-2024-07-18:testleaf-2::DkrC1OVG

🕒 Created At

29 May 2026 18:27:42

📊 Trained Tokens

10,658

🔄 Epochs

2

📋 Batch Size

6

📈 LR Multiplier

1

🔗 Seed

1308033371

📄 Training File

file-RoNpW8xEWfBKovUkWFkmJu

📄 Re-run as Job

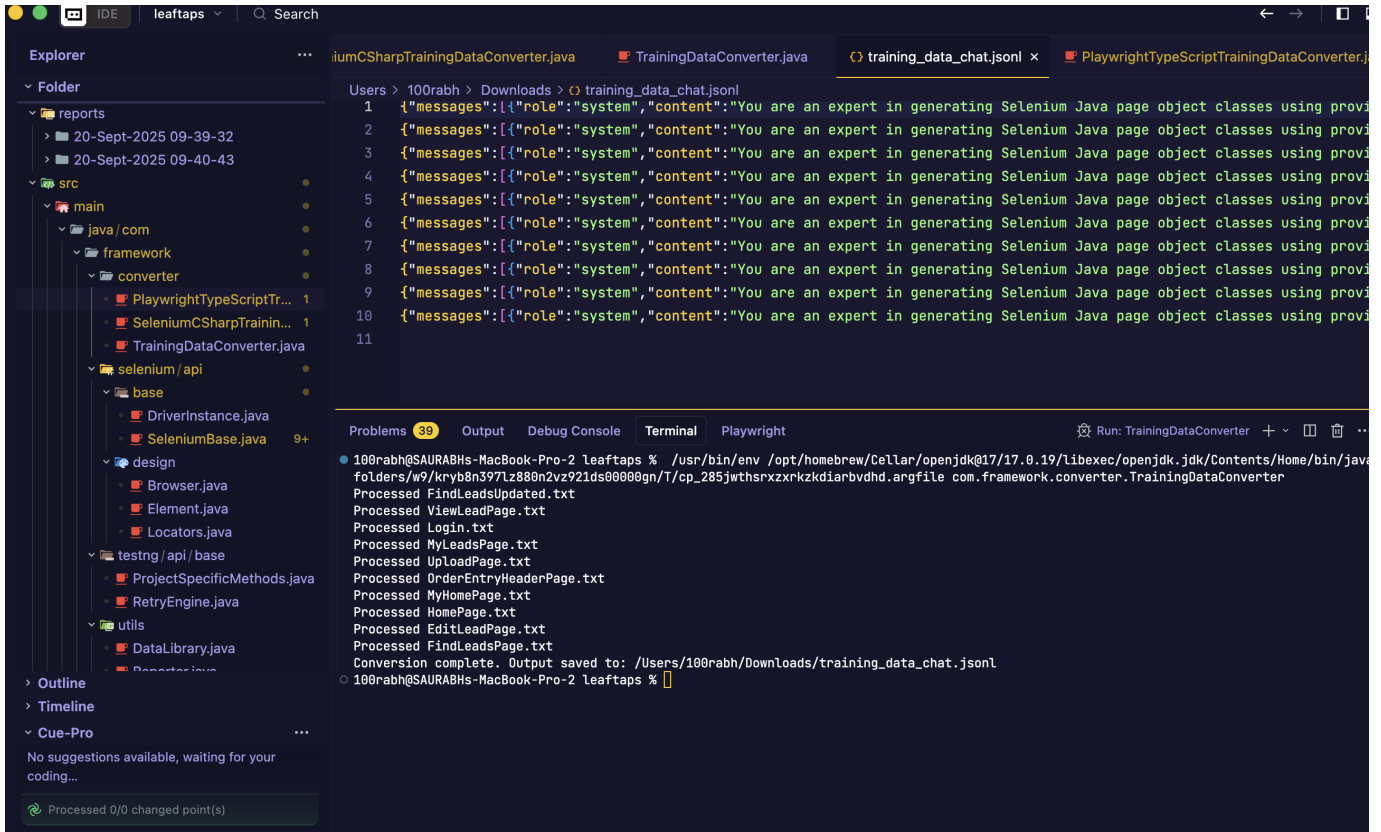
Close

Fine-tuning job completed successfully. The output model is available for integration into the extension.

Fine-tuning Assignment Report

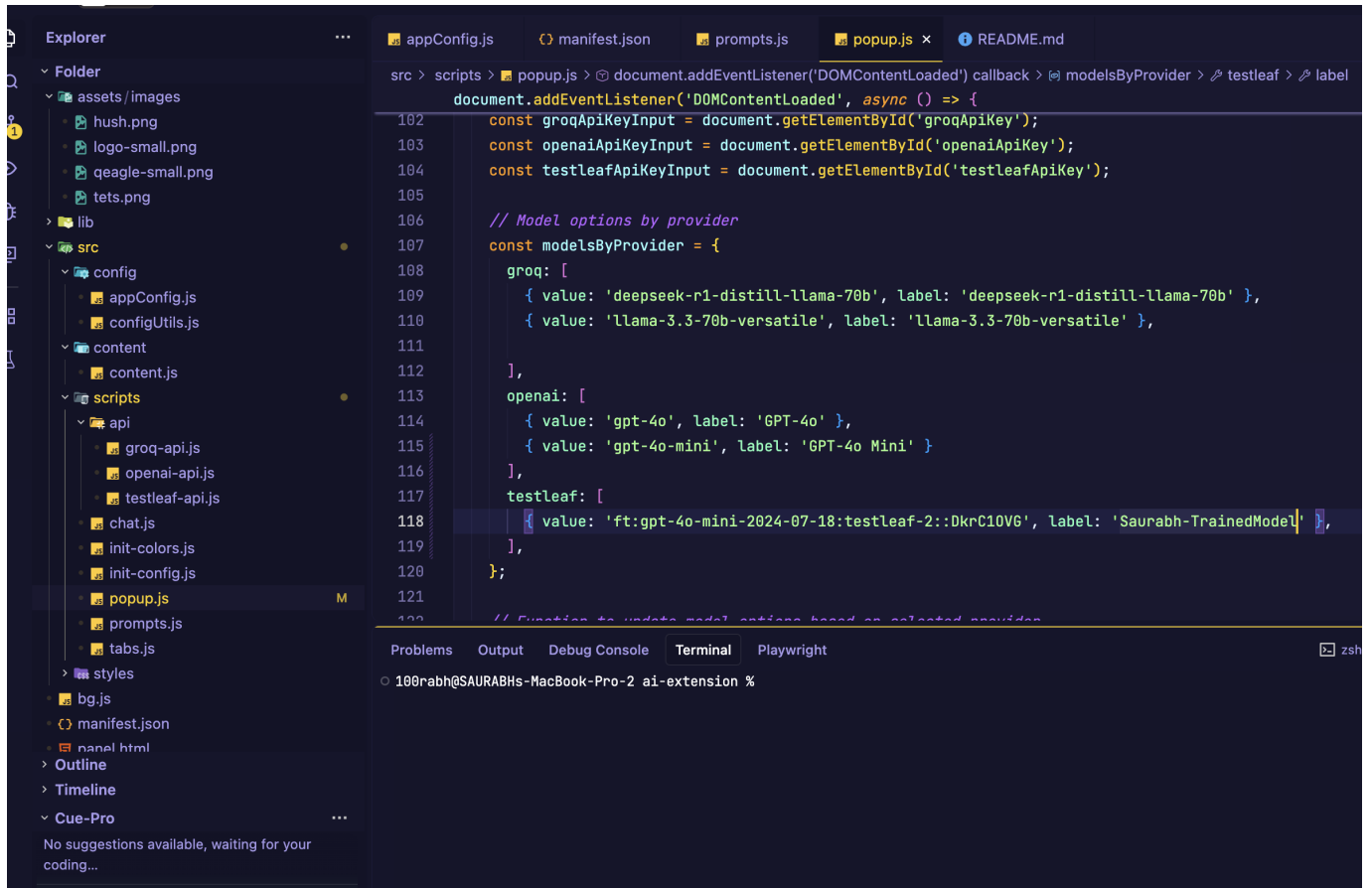
Page 3

Evidence 3



Training data conversion completed. Source page-object examples were converted into chat-format JSONL for supervised fine-tuning.

Evidence 4



```
src > scripts > popup.js > document.addEventListener('DOMContentLoaded', async () => {
  document.addEventListener('DOMContentLoaded', async () => {
102     const groqApiKeyInput = document.getElementById('groqApiKey');
103     const openaiApiKeyInput = document.getElementById('openaiApiKey');
104     const testleafApiKeyInput = document.getElementById('testleafApiKey');
105
106     // Model options by provider
107     const modelsByProvider = {
108       groq: [
109         { value: 'deepseek-r1-distill-llama-70b', label: 'deepseek-r1-distill-llama-70b' },
110         { value: 'llama-3.3-70b-versatile', label: 'llama-3.3-70b-versatile' },
111       ],
112       openai: [
113         { value: 'gpt-4o', label: 'GPT-4o' },
114         { value: 'gpt-4o-mini', label: 'GPT-4o Mini' }
115       ],
116       testleaf: [
117         { value: 'ft:gpt-4o-mini-2024-07-18:testleaf-2::DkrC10V6', label: 'Saurabh-TrainedModel' },
118       ],
119     ];
120   };
121
122   // Function to update model options based on selected provider
```

Fine-tuned model integrated into the extension codebase inside popup.js under the Testleaf provider model options.

Final Result

The assignment requirements were completed: the fine-tuning workflow was executed, the fine-tuned model was created successfully, the output model was configured in **popup.js**, and the AI browser extension was tested by generating Selenium Java Page Object code from an inspected web page.